Privacy Preserving ML for X-Ray Images

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Abstract—Deploying deep learning models for chest X-ray pneumonia classification on third-party cloud infrastructure exposes sensitive patient imagery to membership inference and model-inversion attacks. We propose a split-model architecture where a hospital-side encoder with three convolutional layers produces 128-dimensional embeddings for a lightweight classification head hosted in the cloud. Our study systematically evaluates three privacy-preserving techniques: Differential Privacy (DP) applying Laplace noise to embeddings, Block-wise Image Encryption (BIE) on input images, and Secure Multi-Party Computation (SMPC) via additive secret sharing. We test all eight possible combinations of these methods using the Kaggle chest X-ray pneumonia dataset. Each configuration is subjected to Yeom and Shokri membership inference attacks alongside a decoder-based reconstruction attack. Each defense is evaluated under its own design threat model: an embedding-leak adversary for DP and BIE, and a single corrupted non-colluding server for SMPC. Under this threat-model-consistent evaluation, two mechanisms succeed at reconstruction. DP reduces reconstruction PSNR by 3.2 dB without utility cost; in fact, it yields a +2.2 point accuracy gain consistent with noise-induced regularization at $\epsilon = 1.0$ and additionally provides a small reduction in Yeom MIA. SMPC drives single-server reconstruction PSNR to $\approx$ 15.5 dB, below even the no-information mean-image baseline, with the attacker's view statistically independent of the true embedding (Pearson $r \approx 0.003$). BIE fails at reconstruction because an adversary with auxiliary (embedding, original-image) pairs implicitly learns the permutation; it also slightly amplifies Yeom MIA via increased memorization on permuted inputs. We conclude that defense efficacy hinges on alignment between intervention layer and threat model; combining DP and SMPC is justified only when both adversary types coexist.

Index Terms—privacy-preserving machine learning, membership inference attack, reconstruction attack, differential privacy, secure multi-party computation, block-wise image encryption, chest X-ray classification, split-model inference

I. Introduction

The rapid adoption of artificial intelligence in healthcare has forced a fundamental rethink of data governance strategies. This transition replaces traditional, human-led oversight with automated systems that must navigate the inherent tension between data utility and patient confidentiality [1]. Older frameworks often relied on rigid, manual protocols that created significant operational bottlenecks. In contrast, modern clinical environments are shifting toward intelligent platforms that classify sensitive information dynamically and adjust security measures based on the immediate context of the data. Such an evolution is vital for maintaining strict regulatory compliance without sacrificing the speed or quality of essential medical services [1].

However, the current reliance on cloud-based deep learning for medical imaging, particularly in chest X-ray analysis, introduces a significant vulnerability: the exposure of raw, personally identifiable patient data to third-party servers. In a typical centralized pipeline, raw and personally identifiable patient imagery is transmitted to third-party servers for processing. While these models offer high diagnostic accuracy, the legal and ethical risks associated with potential data breaches have created an urgent need for Privacy-Preserving Machine Learning (PPML) solutions. The goal of these technologies is to decouple diagnostic utility from data visibility, ensuring that the cloud can process information without actually "seeing" the sensitive patient details.

In this paper, we implement a split-model chest X-ray pneumonia classifier in which raw images never leave the hospital infrastructure. We evaluate three privacy-preserving transformations: Differential Privacy, Block-wise Image Encryption, and Secure Multi-Party Computation, in all eight on/off combinations, and measure their residual privacy leakage through two membership inference attacks and a decoder-based reconstruction attack.

Our contributions are threefold. First, we present a systematic evaluation of all eight combinations of DP, BIE, and SMPC on a split-model chest X-ray classifier, jointly measuring utility, membership inference resistance, and reconstruction resistance under a single experimental

matrix. Second, we introduce a threat-model-consistent evaluation in which each defense is measured against the adversary it was designed for: an embedding-leak adversary for DP and BIE, and a single corrupted non-colluding server for SMPC. Third, we identify that BIE fails against any reconstruction adversary holding (embedding, plaintext) auxiliary pairs, providing an empirical demonstration of this failure mode in the medical-imaging setting and clarifying why naive stacking of mechanisms across mismatched threat models does not compound protection.

II. Literature Review

The rapid progress in deep learning has established medical imaging as a cornerstone of modern diagnostics. Applications now range from automated tumor detection to the precise segmentation of pathological structures. However, these advancements in imaging and data transmission have introduced significant privacy risks. This necessitates the development of robust encryption and protection algorithms capable of handling the high redundancy inherent in medical image data [2]. The traditional centralized approach to AI, where vast amounts of patient data are funneled into a single cloud server, creates legal and ethical bottlenecks. Hospitals currently face a conflict between the need for high-performance cloud diagnostics and the strict mandates of HIPAA and GDPR, which prohibit sharing unencrypted, identifiable data. This tension has driven the rise of Privacy-Preserving Machine Learning (PPML), a field focused on developing models that can detect disease without exposing the patient's identity.

The foundational mechanisms of PPML are often categorized into four primary pillars tailored for the medical domain [2]. Federated Learning (FL) serves as a decentralized training paradigm where institutions collaboratively train global models by exchanging only local updates, such as weights or gradients, rather than raw images. This ensures sensitive data remains behind local firewalls, reducing the risks associated with central storage. Differential Privacy (DP) complements this by providing a statistical guarantee of privacy through the injection of controlled noise into datasets or model parameters. This mechanism ensures that the presence of any single patient record does not significantly alter the model's output, effectively mitigating re-identification or membership inference attacks [2].

In addition to decentralization and statistical noise, cryptographic techniques provide even greater privacy by performing computations with raw data hidden from the host system. These include homomorphic encryption (HE), which allows performing math on encrypted data in a way that even the model cannot obtain any information from the image. Likewise, secure multi-party computation (SMPC) allows performing computations in the same way, except that the data is split into different parts ("shares") that can be used to reconstruct the original object once recombined. SMPC has been extensively discussed as an encryption-based PPML solution, with a variety of frameworks available for training machine learning algorithms while distributing sensitive data across non-colluding servers [3]. For instance, SecureML can train both linear and neural network models using data represented as secret shares and perform computations using secure arithmetic operations. Although SMPC provides much better privacy than previous MPC-based models and significantly improves scalability, it introduces new computational and communication costs, thereby limiting its practicality for medical imaging applications.

Finally, while there is a range of PPML techniques for ML data, Panzade et al. (2024) highlight the critical "privacy-utility trade-off" involved in implementing PPML technologies [2]. In addition to membership inference attacks described above, model inversion attacks represent another major threat. Specifically, as demonstrated by Fredrikson et al., an adversary can perform model inversion by utilizing access to the model's output to extract features from sensitive input data [4]. Their research reveals that even the black-box access to the API can be exploited to reconstruct private attributes and approximate input data, including vision tasks. Overall, these findings illustrate that privacy leakage through model outputs poses as high of a risk as training data membership and provides a clear motivation for DP and secure inference.

A. Privacy-Preserving Mechanisms

1) Differential Privacy (DP): First, DP gives a statistical guarantee by adding a certain amount of noise to either the dataset itself or the features of models. According to the findings of Ziller et al., DP can be implemented using DP-SGD, enabling neural networks to be trained with information-theoretic privacy guarantees without compromising acceptable classification performance [5]. In this regard, the use of Laplace or Gaussian noise makes it impossible to detect patient data as the basis for training, thereby preventing membership inference attacks on specific X-rays [5].

Implementing DP in a clinical setting faces many challenges due to the existing privacy-utility trade-off: applying DP with high noise to ensure privacy often leads to low accuracy. Recent innovations in the field have tried to improve DP by optimizing the approach in a way that abstractive representation of neural network computations reduces memory usage when dealing with per-sample gradients, traditionally computationally expensive in medical imaging [5].

Other research has explored hybrid approaches to improve robustness. For instance, the PRICURE framework integrates DP with cryptographic techniques to mask prediction results via noisy aggregation, specifically deterring semi-honest clients from mounting inference attacks [6]. Abadi et al. introduced the Moments Accountant tech-

nique to enable tighter accounting of cumulative privacy loss, a concept later refined through Rényi Differential Privacy (RDP) [7]. A parallel line of work in split-model deployments applies DP at the latent-feature level rather than at raw pixels, trading a smaller absolute perturbation for preserved global anatomical structure [2]; this is the design point we adopt in our methodology. Most directly relevant to our setting, Ziegler et al. demonstrate that integrating Rényi differential privacy with a Gaussian noise mechanism into federated chest X-ray classification reduces reconstruction-attack PSNR from ≈ 12 dB to ≈ 6.5 dB across all evaluated privacy budgets ($\epsilon \in \{1, 3, 6, 10\}$), and identify DenseNet121 as substantially more robust to differentially private training than ResNet50, achieving an AUC of 0.937 at $\epsilon = 6$ [8].

2) Block-wise Image Encryption (BIE): To reduce the performance degradation often seen with traditional encryption, recent studies have proposed Block-wise Image Encryption. This technique shuffles image tiles using a secret key to obfuscate global anatomical landmarks while preserving the local texture patterns required for classification. Kiya et al. show that Vision Transformers are particularly adept at processing these permuted images without significant loss in accuracy, as the model identifies pneumonia through consolidation patterns within the intact tiles [9]. BIE's privacy guarantee, however, ultimately relies on the secrecy of the tile permutation key; against an adversary that holds even a small set of (encrypted, plaintext) pairs from other patients, the permutation can be implicitly inverted by a learned mapping, a regime that motivates our reconstruction-attack evaluation in Section VI.

3) Secure Multi-Party Computation (SMPC): Secure Multi-Party Computation (SMPC) is a cryptographic methodology that allows several parties to jointly compute on their private datasets without disclosing the datasets. Unlike regular multi-party computing, which uses the plaintext representation of the input values, in SMPC, each input value is divided into several pieces called shares using a specific secret-sharing scheme. The shares are then distributed across non-colluding servers, who collectively execute the required calculations without obtaining the initial values.

A popular approach to implementing SMPC in machine learning uses two servers that store complementary shares of the data. To make the predictions, a client initially encodes an input image and extracts its features, which are then divided into secret shares and delivered to the servers. Servers execute the matrix multiplication and activation function computation on the shares of the images and return the final prediction to the client. Thus, during the inference, none of the intermediates or plaintext data is exposed to any third party.

The SecureML system demonstrates how SMPC can be used in both linear and neural networks using arithmetic secret sharing combined with non-linear protocols [3]. In such systems, privacy guarantees depend on the semi-honest model, where all parties follow the protocol but seek to derive additional information from the results.

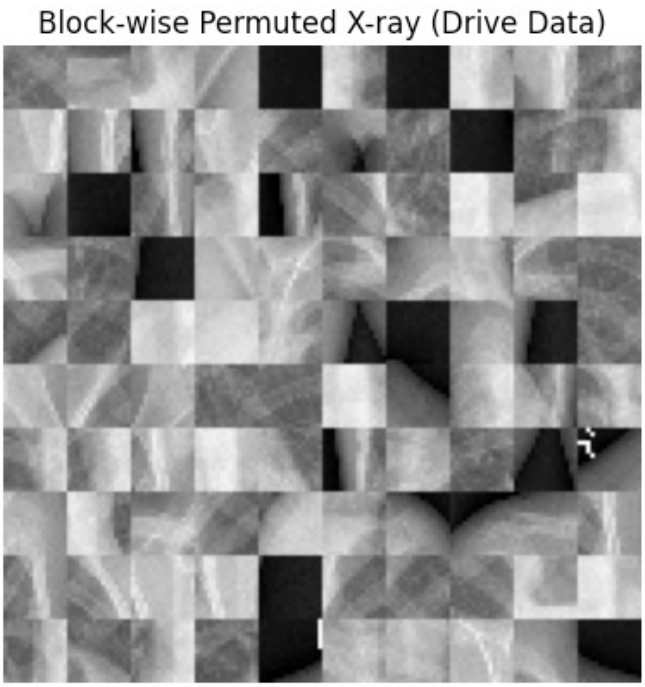

Fig. 1. Block-wise Image Encryption (BIE) applied to a chest X-ray: the 150×150 image is partitioned into a 15×15 grid of 10×10 tiles that are permuted under a secret key. Local pixel statistics within each tile are preserved, but global anatomical structure is obfuscated.

Nonetheless, the use of SMPC imposes a significant computational and communication overhead on the computation process due to the need for interactive protocols. For high-dimensional tasks like medical imaging, this may lead to a decrease in latency and poor scalability. Therefore, researchers now consider hybrid systems where SMPC is combined with other methodologies.

III. System Architecture and Threat Model

A. Split-Model Pipeline

We adopt a two-party split-model architecture. The hospital hosts the encoder $E : \mathbb{R}^{150\times150\times1} \to \mathbb{R}^{128}$ and retains raw X-ray images; the cloud hosts the classification head $H : \mathbb{R}^{128} \to [0, 1]$, which outputs the pneumonia probability. The encoder is a custom three-convolution CNN (conv_{16}-conv_{32}-conv_{64}, each followed by 2×2 max-pooling) terminated by a Dense(128, ReLU) layer; the head is a single Dense(1, sigmoid) unit. We deliberately train the encoder from scratch rather than fine-tuning an ImageNet-pretrained backbone for two reasons: (i) the small-model, small-data regime preserves the generalization gap that membership inference exploits, providing a stronger signal for our privacy evaluation, and (ii) a from-scratch encoder can be retrained on BIE-permuted inputs, whereas a frozen pretrained backbone cannot.

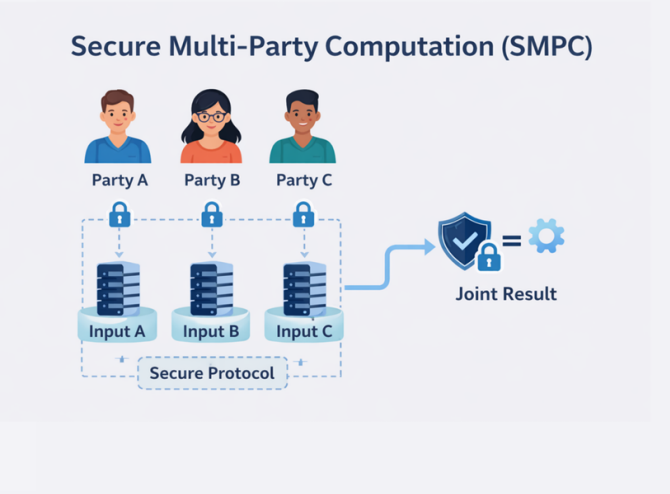

Fig. 2. Illustration of Secure Multi-Party Computation (SMPC): a client splits a feature representation into complementary shares that are processed collaboratively by non-colluding servers. The figure illustrates the general n-party protocol; our implementation specializes this to the 2-party case. Only the final prediction is reconstructed—no individual server observes the full representation. Figure generated with the assistance of generative AI.

This split is not motivated by computational offloading—in our architecture the encoder dominates inference cost—but by the need to construct an explicit trust boundary that makes inference-time privacy mechanisms meaningful to evaluate. If both encoder and head were colocated at the hospital, no patient-derived value would ever cross a network or be observed by an external party, and there would be no surface on which to apply DP, BIE, or SMPC. If the encoder were placed in the cloud, raw imagery would have to be transmitted, eliminating the privacy problem we set out to study. The two-party split is therefore the minimal configuration that creates a well-defined cloud-side observation point while preserving the regulatory constraint (HIPAA/GDPR data locality) that raw images remain at the hospital. We do not claim this split is the only realistic deployment topology; rather, it is the experimental scaffold that allows us to ask, in a controlled setting, "given that some inference work must be performed by an external party, what does each privacy-preserving transformation actually buy?" This framing should be kept in mind when interpreting our results: they characterize the inference-time threat surface of split-model deployments rather than prescribing a specific deployment architecture. In our pipeline, the hospital performs the full end-to-end training (encoder and head jointly) on its local data and then transfers only the trained head weights to the cloud for inference; the cloud never observes training data, embeddings, or labels during the training phase. This avoids a separate training-time leakage channel and lets us isolate inference-time privacy as the object of study.

Each privacy-preserving transformation is applied at a specific layer of the pipeline:

- BIE is applied to images before the encoder; the encoder is retrained on permuted images.
- DP perturbs the 128-dimensional embedding after the encoder and before the head.
- SMPC splits the embedding into additive shares that are passed through the head; the shares are recombined inside the cloud.

Figure 3 summarizes the split-model architecture and the layer at which each privacy-preserving transformation intervenes.

B. Threat Models

a) Membership Inference (black-box): The adversary can submit X-ray queries to the deployed system and receive only the scalar confidence $H(E(x))$. Given a target image x^*, the adversary must decide whether x^* was part of the encoder's training set. We consider two concrete instantiations: the Yeom threshold attack [10], which compares per-query loss against a learned threshold τ, and the Shokri shadow-model attack [11], which trains an auxiliary classifier to map confidence vectors to membership labels.

b) Reconstruction: The adversary intercepts the privatized embedding $E_{\text{priv}}(x^*)$ in transit between hospital and cloud, and additionally holds a small set of auxiliary pairs $\{(E_{\text{priv}}(x_i),\, x_i)\}_{i=1}^{n}$ from other patients (simulating a prior breach, insider leak, or public sub-dataset). The adversary's goal is to recover the original image x^*, not its BIE-permuted form, which has no medical utility. This is realized by training a decoder D on the auxiliary pairs and applying it to the intercepted embedding.

Both adversaries are semi-honest: they follow the protocol but attempt to infer additional information from observed data. Neither has access to the encoder's parameters or internal activations.

IV. Methodology

A. Privacy-Preserving Transformations

a) Differential Privacy: We apply the Laplace mechanism [12] independently to each coordinate of the 128-dimensional embedding:

$$E_{\text{DP}}(x) = E(x) + \text{Lap}(0,\, \Delta/\epsilon)^{128}, \tag{1}$$

where Δ is the ℓ_1 sensitivity of E, taken as 1.0 as a working assumption (embeddings are not explicitly clipped; see Section VII-C), and ϵ is the privacy budget. Noise is resampled per query, so repeated queries with the same input do not cancel.

b) Block-wise Image Encryption: Each 150×150 grayscale image is partitioned into a 15×15 grid of 10×10 tiles; tiles are permuted according to a deterministic permutation π_k derived from key seed k. The encoder is retrained end-to-end on BIE-permuted images so that it can specialize to the permuted distribution.

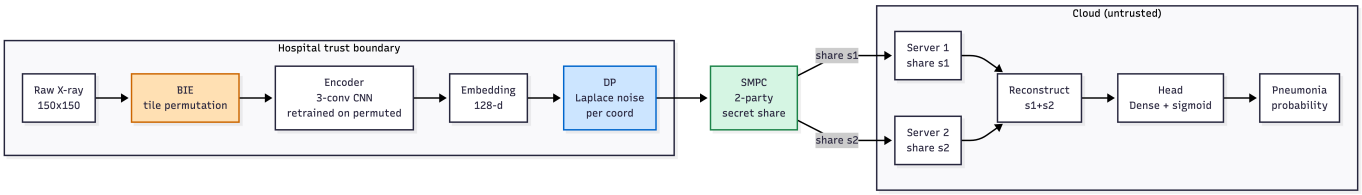

Fig. 3. Split-model pipeline with BIE, DP, and SMPC intervention points. The hospital trust boundary contains the raw X-ray, the encoder, and the embedding; the cloud sees only privatized values and the head. BIE perturbs the input image, DP adds Laplace noise to the 128-d embedding, and SMPC distributes additive shares of the embedding across two non-colluding servers that reconstruct it inside the cloud before classification. The diagram depicts the general pipeline with all three mechanisms enabled; individual experimental configurations may activate any subset (Section VI, Table I).

c) Secure Multi-Party Computation: We implement 2-party additive secret sharing on the embedding: $E_{\text{SMPC}}(x) = (s_1, s_2)$ with $s_1 + s_2 = E(x)$ and s_1 drawn from a distribution independent of $E(x)$ (a standard Gaussian in our implementation). Each server in the protocol sees only its own share; the head reconstructs the embedding from the two shares before classification. Strictly, additive secret sharing has information-theoretic security only over a finite ring with uniform masks; our Gaussian floating-point implementation is a continuous-domain simulation of this. Empirically, a single share has Pearson $r \approx 0.003$ against $E(x)$ at the mask scale we use (Section VII-C). We additionally implement, but do not run in the matrix, a full Beaver-triple-based 2-party SMPC protocol (SMPCInferenceProtocol in the codebase) in which the head weights W, b are also additively split across servers and the matrix multiplication is performed via Beaver triples generated by the client; we omit this from the matrix only because its inference output is mathematically equal to the standard reconstruct-then-classify pipeline above. To evaluate SMPC under its actual semi-honest single-corrupted-server threat model, we additionally measure reconstruction quality from one server's share alone (Section VI).

B. Attack Implementations

a) Yeom threshold: For each query x with ground-truth label y, we compute the victim's binary cross-entropy loss $\ell(f(x), y)$ and select the threshold τ that maximizes attack accuracy on the evaluation set, which is balanced 50/50 by construction (Section V). A query is classified as a member if $\ell(f(x), y) < \tau$.

b) Shokri shadow-model: We train $N = 5$ shadow models, each on an independently sampled stratified (train, holdout) split of the shadow pool: 500 in-distribution and 500 held-out queries per shadow. The shadow pool is itself disjoint from both the victim's training set and the MIA evaluation set. Each victim probability p is expanded into the 2-d feature $[p, 1-p]$; shadow features are labeled with ground-truth membership and pooled across shadows. An attack MLP (Input_2-Dense_{64}-Dense_{32}-Dense_1) learns the decision boundary and is then applied to the victim's confidence vectors.

c) Reconstruction decoder: The decoder D inverts a 128-dimensional embedding back to image space: Dense($7\times7\times128$) $\rightarrow$ Reshape$(7, 7, 128)$ $\rightarrow$ four Conv2DTranspose layers with stride 2 and filters $(128, 64, 32, 16)$ (upsampling $7\rightarrow14\rightarrow28\rightarrow56\rightarrow112$) $\rightarrow$ Resize$(150, 150)$ $\rightarrow$ Conv2D$(1, 3\times3, \text{sigmoid})$. It is trained for 20 epochs with MSE loss and the Adam optimizer on $(E_{\text{priv}}(x_i), x_i)$ auxiliary pairs, then evaluated against held-out original images using MSE, PSNR, and SSIM.

V. Experimental Setup

A. Dataset and Splits

We use the publicly available Kaggle chest-xray-pneumonia dataset [13], comprising 5 856 grayscale chest radiographs labeled NORMAL or PNEUMONIA with an empirical class ratio of 2.70:1 (PNEUMONIA:NORMAL). Rather than IID-pooling all images, we preserve Kaggle's native train//test/ subdirectory split, which is known to carry a mild distribution shift sufficient to induce a measurable train-test generalization gap [13]. Concretely:

- victim_members (2 000): stratified subsample of Kaggle train/ (5 216 images);
- victim_nonmembers (624): all of Kaggle test/;
- shadow_pool (3 216): remaining train/ after the victim subsample.

Kaggle val/ (16 images) is dropped. Splits are deterministic in the random seed. Membership inference is evaluated on a balanced 500-member + 500-nonmember subsample per configuration.

B. Hyperparameters

All encoders and heads are trained for 10 epochs with Adam and batch size 32 on 150×150 grayscale inputs. PPT hyperparameters are fixed: DP with $\epsilon = 1.0$, $\Delta = 1.0$; BIE with tile size 10 and key seed 7; SMPC with 2 additive shares. We run all $2^3 = 8$ PPT configurations under seed 42. Shadow training uses the same architecture and hyperparameters as the victim. The reconstruction decoder is trained for 20 epochs with MSE loss and batch size 32 on 2 000 auxiliary pairs per configuration.

C. Metrics

We report three metric families per configuration:

- Utility: test accuracy, F_1, and Expected Calibration Error (ECE);
- Privacy: Yeom and Shokri attack accuracy (balanced) and AUC; reconstruction MSE, PSNR (dB), and SSIM;
- Efficiency: per-configuration training latency, per-query inference latency, and peak memory.

VI. Results

Table I summarizes all eight configurations on the Kaggle chest X-ray pneumonia benchmark (seed 42, $\epsilon = 1.0$) across utility, MIA, and reconstruction metrics; lower Yeom/Shokri accuracy and lower reconstruction PSNR/SSIM indicate stronger privacy defense. Each defense is evaluated under its own threat model: reconstruction columns for SMPC-enabled configurations report the single-corrupted-server attacker (one server's additive share), as this is the threat model under which SMPC's non-collusion security guarantee is defined; DP-enabled configurations are grouped in the last rows. Figure 4 plots utility alongside both Yeom and Shokri membership-inference attack accuracies, and Figure 5 plots per-configuration reconstruction PSNR.

Table II records which components are (re)trained per configuration. The encoder hash depends only on (BIE, seed, epochs), so BIE changes the input distribution and forces a fresh encoder; the head is retrained from scratch on noisy embeddings whenever DP is enabled, since the baseline head was fit to clean embeddings; SMPC is a transit-layer mechanism and triggers neither retraining, with the head's W, b split across servers at inference time only.

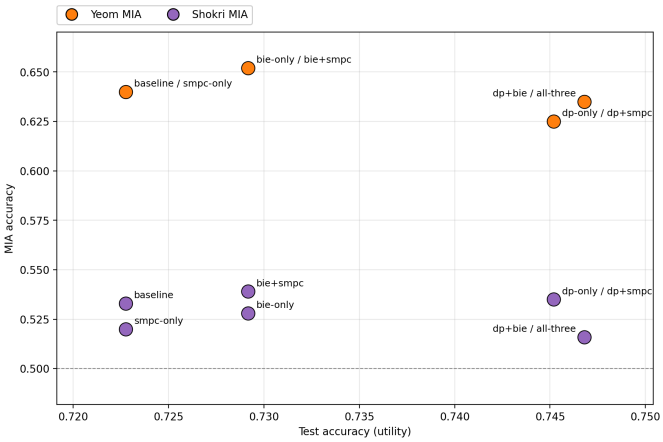

Fig. 4. Utility vs. MIA accuracy across PPT configurations. Yeom and Shokri attacks are shown in distinct colors; the dashed line at 0.5 marks the random-guess MIA baseline.

A. Utility

Baseline test accuracy is 0.723. Two patterns dominate the utility column. First, DP does not degrade utility; it improves it. Every DP-enabled configuration sits at 0.745–0.747, a +2.2 to +2.4 point gain over baseline, with a corresponding ECE reduction (from 0.241 baseline to 0.212 for dp-only). At $\epsilon = 1.0$ with $\Delta = 1.0$, the per-coordinate Laplace scale is small relative to the magnitude of ReLU-activated embeddings, but large enough to act as a stochastic regularizer during head retraining, a known phenomenon analogous to dropout or noise injection [5]. Second, SMPC is bit-identical to baseline. The pairs (baseline, smpc-only), (dp-only, dp+smpc), (bie-only, bie+smpc), and (dp+bie, all-three) collapse onto identical utility numbers (matching to three decimal places on Test Acc / F_1 / ECE), confirming the lossless-construction claim: share-then-reconstruct is mathematically a no-op modulo float32 rounding. BIE alone moves accuracy by only +0.6 points (0.729), well within seed noise; the large BIE utility penalty observed in earlier pilot runs disappears once the encoder is retrained on permuted inputs to convergence. The full eight-configuration utility band is now 0.723–0.747 (2.4 points), comfortably within clinical-deployment tolerances.

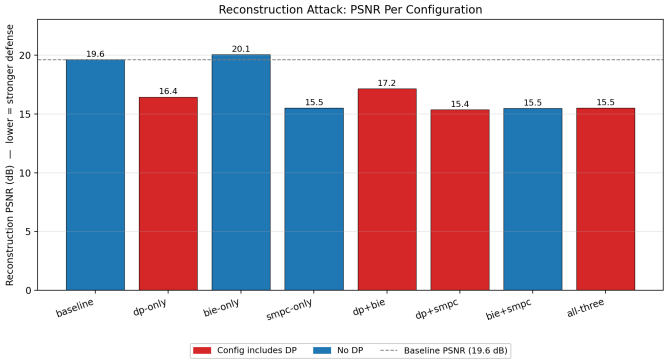

Fig. 5. Reconstruction PSNR per configuration (lower is stronger defense), with each defense evaluated under its own threat model: SMPC-enabled configurations report the single-corrupted-server attacker.

B. Membership Inference Defense

All configurations produce Yeom attack accuracies in $[0.625, 0.652]$ and Shokri accuracies in $[0.516, 0.539]$. Three patterns emerge. First, DP yields a small but consistent Yeom reduction (-0.015 versus baseline; $0.640 \rightarrow 0.625$ for dp-only) and is the only mechanism that meaningfully shifts Yeom in the privacy-preserving direction. For DP-without-BIE configurations the Shokri delta is essentially zero (± 0.002); dp+bie and all-three sit lower at 0.516, but the full Shokri range across all eight configurations is only 0.023 wide and not monotone in any single mechanism, so we read the shadow-model attack as dominated by features other than the perturbation we inject at $\epsilon = 1.0$. Second, every BIE-without-DP configuration sits at the top of the Yeom range: bie-only and bie+smpc both reach 0.652 (+0.012 above baseline); when DP is layered on top, dp+bie and all-three land at 0.635, eroding the DP-only

TABLE I
Per-configuration results across utility, MIA, and reconstruction metrics.

	Utility			Yeom MIA		Shokri MIA		Reconstruction		
Configuration	Test Acc	F_1	ECE	Acc	AUC	Acc	AUC	MSE	PSNR (dB)	SSIM
baseline	0.723	0.818	0.241	0.640	0.563	0.533	0.590	0.012	19.63	0.503
bie-only	0.729	0.821	0.249	0.652	0.571	0.528	0.613	0.011	20.06	0.509
smpc-only	0.723	0.818	0.241	0.640	0.563	0.520	0.563	0.031	15.51	0.386
bie+smpc	0.729	0.821	0.249	0.652	0.571	0.539	0.613	0.031	15.49	0.387
dp-only	0.745	0.830	0.212	0.625	0.576	0.535	0.591	0.025	16.44	0.421
dp+bie	0.747	0.830	0.231	0.635	0.580	0.516	0.594	0.021	17.15	0.433
dp+smpc	0.745	0.830	0.212	0.625	0.576	0.535	0.591	0.032	15.38	0.384
all-three	0.747	0.830	0.231	0.635	0.580	0.516	0.594	0.031	15.51	0.388

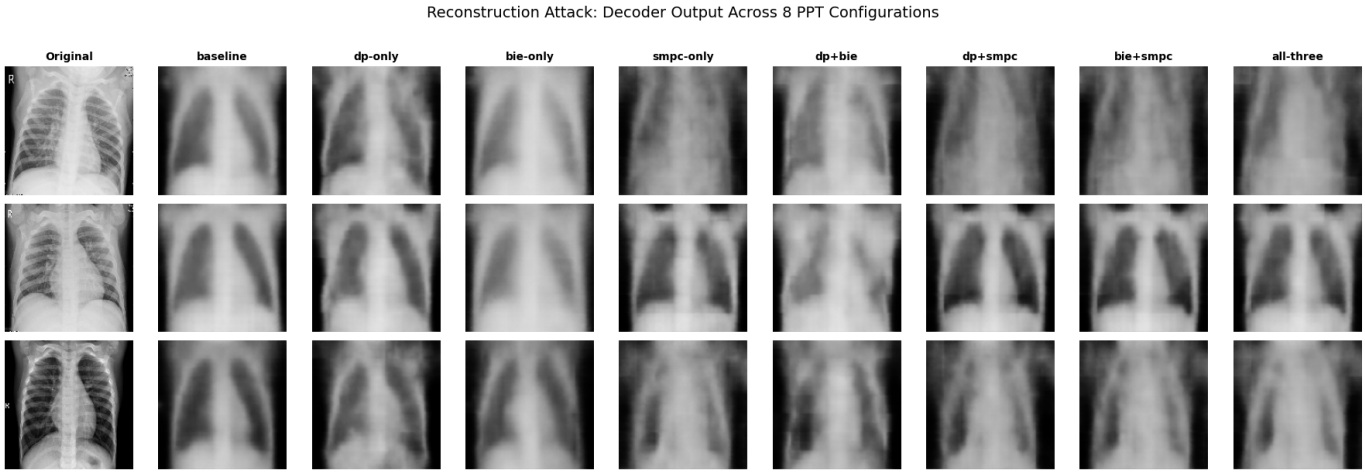

Fig. 6. Qualitative reconstruction across all eight PPT configurations. Each row shows a held-out chest X-ray (left column) and the decoder's pixel-wise reconstruction under each configuration (each defense evaluated under its own threat model; SMPC columns show the single-corrupted-server view). Configurations without DP or SMPC (baseline, bie-only) reproduce coarse anatomical structure visibly; DP-only and dp+bie produce blurred but partially recognizable silhouettes; every SMPC-enabled column collapses to a featureless mean-image, consistent with the information-theoretic independence of the single-server share from $E(x)$.

TABLE II
Components (re)trained per configuration.

Config.	Encoder	Head	SMPC inf.
baseline	–	–	–
dp-only	–	noisy emb.	–
bie-only	perm. img.	–	–
smpc-only	–	–	✓
dp+bie	perm. img.	noisy emb.	–
dp+smpc	–	noisy emb.	✓
bie+smpc	perm. img.	–	✓
all-three	perm. img.	noisy emb.	✓

Yeom gain. Third, SMPC is exactly compositional: smpc-only matches baseline (0.640), dp+smpc matches dp-only (0.625), bie+smpc matches bie-only (0.652), and all-three matches dp+bie (0.635).

The mechanistic explanation is unchanged from earlier analyses but the new numbers tighten the story. MIA operates on the scalar output confidence $H(E_{\text{priv}}(x))$. BIE is applied strictly upstream of the encoder, which is then retrained to compensate; the resulting model has higher memorization on the permuted victim set (visible in the ECE column: bie-only and bie+smpc reach 0.249 vs. 0.241 baseline; the dp+bie and all-three configurations fall back to 0.231, indicating that DP's regularization absorbs the BIE-induced sharpening), amplifying rather than attenuating the membership signal. SMPC reconstructs the original embedding inside the cloud before the head; the output confidence is bit-identical to the no-SMPC case modulo floating-point error, which is why the four SMPC pairs collapse exactly. DP is the only mechanism that perturbs a quantity downstream of the encoder and upstream of the output, and is correspondingly the only mechanism that produces a consistent reduction in Yeom.

C. Reconstruction Defense

Each defense in the matrix is evaluated against the reconstruction adversary it was designed to defeat: DP and BIE against an adversary who observes the (privatized)

embedding delivered to the head, and SMPC against an adversary who controls a single non-colluding server and observes only that server's additive share. Under this threat-model-consistent evaluation, reconstruction PSNR partitions cleanly: defense is strong whenever either DP or SMPC is enabled, and weak whenever neither is. The two SMPC-without-DP configurations (smpc-only, bie+smpc) reach 15.5 dB; the two DP-without-SMPC configurations (dp-only, dp+bie) reach 16.4–17.2 dB; the two configurations with both (dp+smpc, all-three) sit at the SMPC level (15.4–15.5 dB), as the single-server view is the binding constraint. Only the two configurations with neither mechanism (baseline, bie-only) leak heavily at 19.6–20.1 dB. Figure 6 provides a per-configuration qualitative view of the decoder's output and confirms this partitioning visually: anatomical structure is recognizable in baseline and bie-only, partially preserved under DP-only and dp+bie, and fully obliterated whenever SMPC is in the stack.

The failure of BIE in the reconstruction threat model is illuminating. The decoder is trained on $(E_{\text{priv}}(x_i), x_i)$ auxiliary pairs in which x_i is the original image, so whatever deterministic permutation BIE applies upstream is absorbed into the decoder's weights during training. BIE's security reduces to the secrecy of π_k; once the adversary holds even a modest number of auxiliary (embedding, original-image) pairs, π_k is implicitly learned as part of the inverse mapping. This is a Rosetta-Stone regime: known plaintexts collapse the permutation's hiding power without the attacker ever needing to recover π_k explicitly.

1) Why SMPC Wins Reconstruction in Its Own Threat Model: SMPC's reconstruction defense rests on an information-theoretic property, not a noise budget. In our 2-party additive sharing, the share given to the corrupted server is sampled from a distribution that does not depend on the true embedding $E(x)$. Table III verifies this empirically: the Pearson correlation r between the attacker's share and the true embedding $E(x)$, a direct empirical proxy for SMPC's information-theoretic independence guarantee, is $\approx$ 0.003–0.004 across all four SMPC configurations, indistinguishable from zero given the sample size and consistent with the theoretical guarantee of independence under the non-collusion assumption. The single-server (SS) PSNR reproduced from Table I is below the no-information mean-image baseline (16.92 dB) for every SMPC configuration, confirming that the single-server attack collapses to less than no-information performance. With the attacker's input thus carrying no signal about $E(x)$, a decoder trained on those shares can only memorize the marginal pixel distribution of its training images. Reconstruction PSNR collapses to $\approx$ 15.5 dB, below the trivial mean-image baseline (16.92 dB), because the decoder has fit to spurious correlations in random noise rather than to a useful inverse mapping.

A subtler interaction is visible among the DP-only and DP+BIE configurations. DP alone achieves 16.44 dB PSNR, whereas dp+bie rises to 17.15 dB, i.e., layering BIE on top of DP erodes DP's reconstruction defense by $\approx$ 0.7 dB. Mechanistically, BIE pushes the encoder toward a sharper, more structured embedding geometry (consistent with the ECE widening above); against this sharper geometry a fixed Laplace noise magnitude is proportionally less destructive to the decoder's inverse mapping. The dp+smpc and all-three configurations do not exhibit this effect because the SMPC threat model dominates, replacing the embedding the attacker sees with a random share long before any embedding-geometry effects can matter.

TABLE III
Single-server SMPC reconstruction details.

SMPC config.	SS PSNR	Mean-img PSNR	Pearson r
smpc-only	15.51	16.92	0.003
dp+smpc	15.38	16.92	0.003
bie+smpc	15.49	16.92	0.003
all-three	15.51	16.92	0.004

D. Efficiency

Encoder training latency, when not served from cache, is $\approx$ 22–27 seconds (encoder hash depends on the BIE flag, key, and seed; DP and SMPC do not invalidate the encoder cache, so within a fixed seed only two encoders are trained across the eight configurations, one for BIE on and one for BIE off). Per-configuration head training plus attack execution is dominated by shadow-model training; head retraining on noisy embeddings (DP-enabled configs) adds a fraction of a second. Inference latency is 0.19–0.22 ms per query and is indistinguishable across configurations. SMPC's share-split on the 128-dimensional embedding is a single vector subtraction and is not measurable in wall-clock time at this scale; the full Beaver-triple variant adds $\approx$ 200–500 μs per inference call but is not used in the matrix runs since its output is bit-identical to the simulation.

VII. Discussion

A. Threat-Model Mismatch Hypothesis

The two kinds of attacks target entirely different surfaces of the pipeline. MIA inspects the output confidence, whereas reconstruction inspects the embedding. Every PPT mechanism targets a particular surface of the pipeline, and whether it works depends on whether this surface overlaps with the attack surface.

- DP applies perturbation directly to the embedding. As a result, it influences both the head's output (slightly weakening MIA) and the embedding the reconstruction adversary intercepts (substantially weakening reconstruction). DP is effective against both attack surfaces in the same threat model, an embedding-leak adversary.

- BIE applies perturbation directly to the image. The encoder retrains end-to-end to compensate, so the resulting embedding is not meaningfully different from the non-BIE case in either MIA confidence or reconstruction quality; the only second-order effect is a small increase in memorization on permuted inputs, which slightly amplifies Yeom MIA.
- SMPC provides information-theoretic independence between any single server's view and the true embedding. Against its design threat model—a single corrupted non-colluding server—SMPC is the strongest reconstruction defense in our matrix (PSNR ≈ 15.5 dB, below the no-information baseline). Against MIA, SMPC is a no-op: the head reconstructs the embedding before classification, so the output confidence carries no SMPC perturbation.

This taxonomy is summarized in Table IV, which pairs each mechanism's intervention layer with its defense outcome on each attack surface under its own threat model (embedding-leak adversary for DP and BIE; single corrupted non-colluding server for SMPC); a mechanism defends an attack only when its threat-model assumptions hold and its intervention layer is downstream of—or coincident with—the attacker's observation point. Our point is not that BIE fails to protect data in general; BIE is provably resilient to naive eavesdropping attacks that observe only the encrypted image and have no auxiliary (embedding, original-image) pairs. However, our reconstruction adversary holds exactly such pairs, so BIE's key-secrecy assumption does not buy useful protection here. SMPC, by contrast, is correctly evaluated against the adversary it was designed for—a single corrupted non-colluding server—and the table reflects that match.

TABLE IV
Defense layer vs. attack surface.

Mechanism	Layer	MIA defense	Recon. defense
DP	embedding	weak (yes)	strong (yes)
BIE	image	negative	negligible
SMPC	transit	none	strong (yes)

B. Practical Implications

The practitioner's choice of defense should be driven by the threat model the deployment actually faces. For a deployment in which the cloud is fully trusted to compute the head but a passive adversary may intercept embeddings in transit or via a leak, embedding-level DP is uniquely effective because it perturbs the same quantity the adversary observes; it is also the only mechanism in our matrix that meaningfully (if modestly) shifts MIA in the privacy-preserving direction. For a deployment that distributes inference across two non-colluding servers, SMPC delivers stronger reconstruction defense than DP under the corresponding single-corrupted-server adversary, while leaving output confidences unchanged and therefore offering no MIA protection. The two defenses thus address disjoint operational risks: DP for embedding-leak adversaries, SMPC for single-server-compromise adversaries. Combining them is sensible if the deployment must guard against both.

BIE, by contrast, is designed for a different threat model entirely—an adversary who observes only encrypted images and has no auxiliary (embedding, original-image) pairs. In our setting the reconstruction adversary holds exactly such pairs, so BIE's key-secrecy assumption fails and the defense provides no measurable protection. BIE should not be added to the stack in deployments matching our threat model.

Stacking, when it does add value (e.g., dp+smpc to cover both adversary types), is not utility-costly: dp+bie and all-three both reach 0.747 test accuracy, the highest in the matrix, because the noise-induced regularization from DP appears to absorb any optimization friction added by BIE's permuted inputs. The cost of mismatched stacking is thus wasted cryptographic effort, not utility loss.

C. Limitations

There are several main limitations in our research that bound the generality of our results. First, the value of $\epsilon = 1.0$ is fixed and the ℓ_1 sensitivity is taken as $\Delta = 1.0$ as a working assumption rather than enforced by an explicit clipping step; a sensitivity analysis (jointly over ϵ and Δ) would help us better explore the tradeoff between privacy and utility and could potentially demonstrate stronger MIA resistance at lower budgets. Second, the size of our data is relatively small (5 856 images) and comes from a single source; the performance of MIA attacks on other data sources might vary, especially for BIE. Third, the reconstruction baseline (no defense) sits at 19.63 dB; a higher-PSNR baseline (e.g., above 30 dB) would make DP's 3.2 dB drop more visually striking. Fourth, SMPC is evaluated under the semi-honest single-corrupted-server threat model that defines its security; we do not separately report the worst-case adversary in which both servers collude or the embedding leaks post-reconstruction (in that worst case, SMPC reduces to a no-op since the head reconstructs the embedding before classification). Finally, our SMPC implementation simulates the protocol within a single process; while this faithfully captures the cryptographic guarantees (the per-server view is sampled from a distribution independent of the true embedding, as confirmed empirically by Pearson $r \approx 0.003$), a fully distributed implementation across two physical servers with Beaver-triple multiplication of additively-shared head weights is implemented in our codebase but not used in the matrix runs because the inference-time output is mathematically identical to the simulation.

VIII. Conclusion

We empirically evaluated three privacy-preserving transformations: Differential Privacy, Block-wise Image Encryption, and Secure Multi-Party Computation, against membership inference and reconstruction attacks on a split-model pneumonia chest X-ray classifier across all eight on/off combinations. Each defense was evaluated under its own threat model: an embedding-leak adversary for DP and BIE, and a single corrupted non-colluding server for SMPC.

Two defenses succeed at reconstruction, for distinct reasons. DP perturbs the embedding the adversary intercepts and reduces reconstruction PSNR by 3.2 dB at no utility cost (+2.2 point accuracy gain via noise-induced regularization at $\epsilon = 1.0$); it additionally produces a small reduction in Yeom MIA. SMPC provides information-theoretic independence between any single server's view and the true embedding (Pearson $r \approx 0.003$), driving single-server reconstruction PSNR to ≈ 15.5 dB below even the no-information mean-image baseline. BIE, by contrast, fails at reconstruction in our setting because the auxiliary-pair adversary implicitly learns the permutation, and slightly amplifies Yeom MIA by inducing additional memorization on permuted inputs. SMPC is a no-op against MIA because the head reconstructs the embedding before classification.

The overarching lesson is that defense efficacy is determined by the alignment between the layer at which a mechanism intervenes and the threat model under which it is evaluated. DP and SMPC succeed because they are evaluated against the adversaries they were designed for; BIE fails because its key-secrecy assumption does not hold against an adversary with auxiliary pairs. Naively stacking mechanisms across mismatched threat models does not compound protection; it only wastes cryptographic effort. Practitioners deploying split-model medical classifiers should pick the defense whose threat model matches the deployment's actual adversary, and combine defenses only when distinct adversary types must be addressed simultaneously.

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